

Internship Project Assignment – Jean Pierre Ndikumana

Project Title: Team Collaboration & Task Management Hub

Project Objective The objective of this project is to design and develop a real-time team collaboration and task management system. The platform will allow teams to create projects, assign tasks, set deadlines, track progress with live updates, and maintain activity feeds — a lightweight clone of Trello/Asana with WebSocket-powered real-time features.

You must follow a structured development approach starting with system documentation, then implementation using **Node.js + React**. All work must be managed using GitHub.

Phase 1: System Design and Documentation

Before starting development, you must prepare and submit the following:

1. **Software Requirements Specification (SRS)** The SRS must include:
 - System overview
 - User roles and permissions
 - Functional requirements
 - Non-functional requirements
 - System modules
 - Use cases
2. **Data Flow Diagrams (DFDs)** You must provide:
 - Context diagram
 - Level 1 DFD
 - Level 2 DFD where necessary
3. **Entity Relationship Diagram (ERD)** The ERD must clearly define:
 - Entities such as Users, Teams, Projects, Tasks, Comments, Notifications
 - Relationships between entities
 - Primary and foreign keys

Approval of these documents is required before moving to development.

Phase 2: System Development

Technologies to Use

- **Backend:** Node.js + Express.js
- **Frontend:** React.js (with Vite or Create React App)
- **Real-time:** Socket.io (WebSockets)
- **Database:** MongoDB (preferred) or PostgreSQL
- **Authentication:** JWT + bcrypt
- **UI:** Tailwind CSS or Bootstrap + React components

System Overview The system will support the following users:

- **Team Members** – Create/view tasks, update status, comment
- **Project Managers** – Create projects, assign tasks, manage team
- **Admin** – Manage users, teams, and system overview

Core Features

1. **User & Team Management**
 - User registration and login
 - Role-based access control (Admin, Manager, Member)
 - Create and join teams
2. **Project Management**
 - Create projects with boards
 - Kanban-style task boards (To Do, In Progress, Review, Done)
3. **Task Management**
 - Create tasks with title, description, assignee, due date, priority, labels
 - Drag & drop task movement (real-time)
 - Comments and attachments on tasks
4. **Real-time Features**
 - Live board updates using WebSockets (Socket.io)
 - Real-time activity feed
 - Notifications when assigned or status changes
5. **Admin Dashboard**
 - Manage users and teams
 - View all projects and system activity
6. **Reporting**
 - Task completion reports
 - Team productivity overview
 - Due date tracking
7. **Search and Filtering**
 - Filter tasks by status, assignee, priority, due date

Backend Requirements

- RESTful APIs with Node.js + Express
- Proper controllers, services, and models
- Database migrations / schema design
- Real-time event broadcasting with Socket.io
- Input validation and error handling
- File upload support (for task attachments)

Frontend Requirements

- Responsive UI with modern design
- Drag & drop Kanban board
- Real-time updates without page refresh
- Role-based dashboards

- Clean and intuitive task cards

GitHub Version Control Requirements

- Create a GitHub repository
- Use branches: main, dev (and feature branches)
- Commit frequently with clear messages
- Push code continuously during development
- Maintain a comprehensive README with project description, setup instructions, and technologies used

Expected Deliverables

- Approved SRS document
- DFDs and ERD
- Fully working full-stack application
- Real-time functionality with Socket.io
- Complete task workflow and notifications
- Clean, well-structured code

Acceptance Criteria

- Documentation is complete and approved
- Users can log in and perform role-based actions
- Real-time updates work across users
- Drag & drop and notifications function correctly
- System functions without major errors
- Code is clean and properly organized
- GitHub shows continuous development

Project Timeline Maximum: 2 weeks

All progress must be pushed to GitHub continuously during development.